1. All the following are correct about the sheep (Dolly) cloning experiment EXCEPT:
   - a. The cultured cells used were at G2 phase of the cell cycle
   - b. Egg activation was achieved by the electric current
   - c. They used electric current to fuse the mammary gland cell with the enucleated egg
   - d. They used the mammary gland cells as the donor of the genetic material
2. Epithelial adult stem cells in the lining of the digestive tract occur in deep crypts and give rise to the following cell types EXCEPT:
   - a. Entero-endocrine cells
   - b. Goblet cells
   - c. Absorptive cells
   - d. Follicle cells
3. Totipotency of any cell nucleus can be tested by transplanting the nucleus to an enucleated egg of the same species, and any substantial activity can be ascribed to the transplanted nucleus.
   - a. True
   - b. False
4. The difference between therapeutic cloning and reproductive cloning is:
   - a. In therapeutic cloning, the enucleated egg is activated by a sperm, while in reproductive cloning, it is activated by electric shock
   - b. In therapeutic cloning, the embryo is allowed to develop to the blastocyst stage only, while in reproductive cloning, the embryo is allowed to develop into a whole organism after transferring it to a foster mother
   - c. Therapeutic cloning is not ethical, while reproductive cloning is ethical
   - d. The goal of therapeutic cloning is to produce a whole organism, while the goal of reproductive cloning is to produce embryonic stem cells for therapy
5. Skin adult stem cells are usually residing in:
   - a. Basal layer of the epidermis
   - b. Hypodermis
   - c. Dermis layer of the skin
   - d. Inner layer of the dermis
6. Which of these is NOT considered a psychological harm of reproductive cloning in humans?

- a. Loss of uniqueness to the child
- b. To avoid homozygosity in case of recessive alleles (parents are carriers)
- c. Harms to other participants, including egg donors, gestational surrogates, etc.
- d. Loss of self-worth to the DNA donor

7. The ability of adult stem cells to form specialized cell types of tissues other than the place where they reside is known as:
    - a. Transformation
    - b. Transduction
    - c. Translation
    - d. Transdifferentiation
8. The difference between embryonic stem cells and adult stem cells is:
    - a. Adult stem cells are easy to isolate, while embryonic stem cells are difficult
    - b. Embryonic stem cells can differentiate into limited types of cells, while adult stem cells can differentiate into all types of cells
    - c. Adult stem cells are easy to culture, while embryonic stem cells are difficult
    - d. Adult stem cells are rare, while embryonic stem cells are abundant
9. Which step of culturing human embryonic stem cells (ESC) contains a wrong statement?
    - a. The inner surface of the culture dish is typically coated with a feeder layer
    - b. Human embryonic stem cells are isolated by transferring the inner cell mass into a plastic laboratory culture dish that contains a nutrient broth known as culture medium
    - c. Over the course of several days, the ESC proliferate and begin to crowd the culture dish
    - d. The reason for having the mouse feeder cells in the bottom of the culture dish is to give the inner cell mass cells protection from microbes
10. In comparison between Briggs and King's cloning experiment and Gurdon's cloning experiment, which of the following is NOT true?
    - a. Briggs and King used *Rana pipiens* amphibian species, while Gurdon used *Xenopus laevis* amphibian species
    - b. The first successful cloning was achieved by transplanting the nucleus from the intestine of the tadpole into an enucleated egg using *Rana pipiens* species
    - c. The result of both experiments showed that as the age of the transplanted nucleus gets older, the rate of cloning success decreases

- d. Gurdon employed the serial nuclear transfer technique, while Briggs and King did not

11. Embryonic stem cells from a donor introduced into a patient could cause transplant rejection.
    - a. True
    - b. False
12. Diseases that might be treated by transplanting cells generated from human embryonic stem cells include all the following EXCEPT:
    - a. Headache
    - b. Diabetes
    - c. Parkinson's disease
    - d. Traumatic spinal cord injury
13. All the following are examples of adult stem cell plasticity that have been reported during the past few years EXCEPT:
    - a. Brain stem cells may differentiate into blood cells and skeletal muscle cells
    - b. Bone marrow stromal cells may differentiate into cardiac muscle cells and skeletal muscle cells
    - c. Hematopoietic stem cells may differentiate into three major types of brain cells (neurons, oligodendrocytes, and astrocytes), skeletal muscle cells, cardiac muscle cells, and liver cells
    - d. Neural stem cells in the brain give rise to its three major cell types: nerve cells (neurons) and two categories of non-neuronal cells—astrocytes and oligodendrocytes
14. Physical transfection includes all the following EXCEPT:
    - a. Transduction
    - b. Electroporation
    - c. Microinjection
    - d. Particle bombardment (gene gun)
15. Adult human stem cells are found in all the following EXCEPT:
    - a. Lining of the digestive system
    - b. Brain
    - c. Human eyes
    - d. Skin

16. The process of extracting stem cells involves killing the embryo. To many pro-lifers, this is murder. They feel that murdering one person, the embryo, to cure another person of paralysis, diabetes, heart disease, etc., can never be justified.
    - a. True
    - b. False
17. New studies indicate that it may be possible to direct the differentiation of human embryonic stem cells in cell culture to form insulin-producing cells that eventually could be used in transplantation therapy for diabetics.
    - a. True
    - b. False
18. Which of the following organisms' embryos must undergo irradiation to increase the percentage of the transfected cells in the developing embryos?
    - a. Mice
    - b. Cows
    - c. Birds
    - d. Fishes
19. Bone marrow stromal stem cells generate all the following EXCEPT:
    - a. Red blood cells
    - b. Fat cells
    - c. Bone
    - d. Cartilage
20. Which of these is mismatched?
    - a. Totipotent cell — Zygote
    - b. Totipotent cells — Cells of the four-cell stage embryo
    - c. Pluripotent cells — Inner cell mass
    - d. Totipotent cells — Inner cell mass

Q1	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**	**Q11**	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
A	D	B	B	A	B	D	D	D	B	A	A	D	A	C	B	A	C	D	D

21. Which of the following is NOT normally derived from bone marrow stem cells?
    - a. All types of blood cells
    - b. Bone
    - c. Cartilage
    - d. Heart muscle fibers
22. The vector that can be used to transfer 80-90 kb of DNA is:
    - a. Yeast chromosome
    - b. Multicopy plasmid
    - c. Bacteriophage lambda
    - d. Bacteriophage P1
23. All the following are examples of good cloning in humans EXCEPT:
    - a. Providing organ transplants
    - b. Treating infertility
    - c. Producing a large number of identical human individuals
    - d. Replacing a loved one
24. All the following laboratory tests are used to identify embryonic stem cells EXCEPT:
    - a. Growing and subculturing the stem cells for many months to ensure long-term self-renewal
    - b. Inspecting the cultures through a microscope to ensure the cells look healthy and remain undifferentiated
    - c. Using specific techniques to determine the presence of surface markers found only on undifferentiated cells
    - d. Testing the presence of surface markers found on differentiated cells
25. The blastocyst includes all the following parts EXCEPT:
    - a. Blastoderm cells
    - b. Inner cell mass
    - c. Blastocoel
    - d. Trophoblast cells
26. Which of these is NOT a characteristic of adult stem cells?
    - a. The number of adult stem cells is very rare

- b. They are usually quiescent and become active upon injury
- c. It is very difficult to isolate adult stem cells
- d. Culturing adult stem cells is very easy

27. Which of the following organisms' embryos must undergo irradiation to increase the percentage of transfected cells in the developing embryos?
    - a. Mice
    - b. Cows
    - c. Birds
    - d. Fishes
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    - a. Red blood cells
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    - a. Totipotent cell — Zygote
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    - c. Pluripotent cells — Inner cell mass
    - d. Totipotent cells — Inner cell mass
30. The difference between therapeutic cloning and reproductive cloning is:
    - a. In therapeutic cloning, the embryo is allowed to develop to the blastocyst stage only, while in reproductive cloning, the embryo is allowed to develop into a whole organism after transferring it to a foster mother
    - b. In therapeutic cloning, the enucleated egg is activated by a sperm, while in reproductive cloning, it is activated by electric shock
    - c. Therapeutic cloning is not ethical, while reproductive cloning is ethical
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Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
D	D	C	D	A	D	C	D	D	A

31. **Question 31**: Which of the following is NOT a characteristic of adult stem cells?
    - a. The number of adult stem cells is very rare
    - b. They are usually quiescent and become active upon injury
    - c. It is very difficult to isolate adult stem cells
    - d. Culturing adult stem cells is very easy
32. **Question 32**: Which of the following organisms' embryos must undergo irradiation to increase the percentage of transfected cells in the developing embryos?
    - a. Mice
    - b. Cows
    - c. Birds
    - d. Fishes
33. **Question 33**: Bone marrow stromal stem cells generate all the following EXCEPT:
    - a. Red blood cells
    - b. Fat cells
    - c. Bone
    - d. Cartilage
34. **Question 34**: Which of these is mismatched?
    - a. Totipotent cell — Zygote
    - b. Totipotent cells — Cells of the four-cell stage embryo
    - c. Pluripotent cells — Inner cell mass
    - d. Totipotent cells — Inner cell mass
35. **Question 35**: The difference between therapeutic cloning and reproductive cloning is:
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    - b. In therapeutic cloning, the enucleated egg is activated by a sperm, while in reproductive cloning, it is activated by electric shock
    - c. Therapeutic cloning is not ethical, while reproductive cloning is ethical
    - d. The goal of therapeutic cloning is to produce a whole organism, while the goal of reproductive cloning is to produce embryonic stem cells for therapy

36. **Question 36**: Which of the following is NOT normally derived from bone marrow stem cells?
    - a. All types of blood cells
    - b. Bone
    - c. Cartilage
    - d. Heart muscle fibers
37. **Question 37**: The vector that can be used to transfer 80-90 kb of DNA is:
    - a. Yeast chromosome
    - b. Multicopy plasmid
    - c. Bacteriophage lambda
    - d. Bacteriophage P1
38. **Question 38**: All the following are examples of good cloning in humans EXCEPT:
    - a. Providing organ transplants
    - b. Treating infertility
    - c. Producing a large number of identical human individuals
    - d. Replacing a loved one
39. **Question 39**: All the following laboratory tests are used to identify embryonic stem cells EXCEPT:
    - a. Growing and subculturing the stem cells for many months to ensure long-term self-renewal
    - b. Inspecting the cultures through a microscope to ensure the cells look healthy and remain undifferentiated
    - c. Using specific techniques to determine the presence of surface markers found only on undifferentiated cells
    - d. Testing the presence of surface markers found on differentiated cells
40. **Question 40**: The blastocyst includes all the following parts EXCEPT:
    - a. Blastoderm cells
    - b. Inner cell mass
    - c. Blastocoel
    - d. Trophoblast cells

Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
D	C	D	D	A	D	D	C	D	A